

Supplementary Material: Convergent Discovery of Critical Phenomena Mathematics Across Disciplines

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1 Metatron Dynamics Operator Definitions

The Metatron Dynamics framework composes four operators (A , B , R , C) into a composite evolution operator E . All operators act on discrete fields $x \in \mathbb{R}^n$ with circular boundary conditions.

A — Gradient Extraction (symmetric):

$$A(x)[i] = x[i] - \text{mean}(x) \tag{1}$$

Zero-sum ($\sum A(x)[i] = 0$), translation-invariant.

B — Local Coupling (symmetric):

$$B(x)[i] = x[i] + x[(i+1) \bmod n] \tag{2}$$

Couples each element to its forward neighbor under circular boundary conditions.

R — Antisymmetric Circulation:

$$R(x, \rho)[i] = x[i] + \rho(x[(i+1) \bmod n] - x[(i-1) \bmod n]) \tag{3}$$

Parameterized by circulation strength ρ . Breaks equilibrium symmetry; directional flow sustains coherent structures under iteration.

C — Bounded Coherence:

$$C(x)[i] = \frac{x[i]}{1 + |x[i]|} \tag{4}$$

Saturating nonlinearity with bounded output ($|C(x)[i]| \leq 1$), sign-preserving.

E — Composite Evolution:

$$E(x, \rho) = C(R(B(A(x)), \rho)) \tag{5}$$

In the linear regime ($|x| \ll 1$), the operators A , B , R commute; ordering matters only through the position of the nonlinear operator C . Placing C last (after A , B , R) allows the linear operators to build spectral structure before bounded compression.

Criticality Detection via κ_m

The contraction factor κ_m admits two equivalent formulations:

Spectral:

$$\kappa_m = \lambda_{\max}(J_E) \quad (6)$$

where J_E is the Jacobian of E and $\lambda_{\max}(\cdot)$ denotes the largest eigenvalue magnitude (spectral radius).

Geometric (Lipschitz constant):

$$\kappa_m(x) = \lim_{\varepsilon \rightarrow 0} \frac{\|E(x + \varepsilon, \rho) - E(x, \rho)\|}{\|\varepsilon\|} \quad (7)$$

Both yield the same classification:

- $\kappa_m < 1$: contracting (perturbations decay)
- $\kappa_m \rightarrow 1$: critical (perturbations persist, $\tau \rightarrow \infty$)
- $\kappa_m > 1$: expanding (perturbations grow)

2 Correspondence Test: Ising Model

Using the 2D Ising model at $T_c = 2.269$ (Onsager 1944), we tested whether three κ_m candidates—spatial (ξ -based), temporal (τ -based), and composite ($\sqrt{\xi\tau}$)—correctly identify the critical point. Lattice: 32×32 ; 2000 equilibration sweeps; 200 samples per temperature.

Table 1: Metatron Dynamics correspondence test results

T	$\kappa_{m,\text{spatial}} (\xi)$	$\kappa_{m,\text{temporal}} (\tau)$	$\kappa_{m,\text{composite}}$
2.0	2.0	1.34	1.64
2.1	3.0	4.69	3.75
2.2	3.0	2.71	2.85
2.269	5.0	18.60	9.64
2.3	4.0	4.94	4.45
2.4	4.0	14.80	7.69
2.5	4.0	6.35	5.04

All three candidates peak at $T = T_c$. Temporal correlation τ increases $14\times$ at the critical point, demonstrating critical slowing down. The temporal measure shows elevated variance at temperatures near T_c , consistent with finite-size effects on a 32×32 lattice; the secondary elevation at $T = 2.4$ illustrates why larger lattices are needed to establish a clean divergence profile. The 32×32 lattice is illustrative; standard errors are not reported for these 200 samples, and both full statistical characterization and direct computation of $\kappa_m = \lambda_{\max}(J_E)$ from the E operator’s Jacobian require larger lattices.